



US012402711B2

(12) **United States Patent**
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(10) **Patent No.:** **US 12,402,711 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **APPLICATOR FOR APPLYING A PRODUCT, IN PARTICULAR A COSMETIC PRODUCT**

(56) **References Cited**

U.S. PATENT DOCUMENTS

2018/0344001 A1 * 12/2018 Tanaka A45D 33/006
2019/0000212 A1 * 1/2019 Balestrini A45D 34/043
2020/0367631 A1 * 11/2020 Bianco A45D 40/265
2020/0383457 A1 * 12/2020 Park A45D 40/265

FOREIGN PATENT DOCUMENTS

EP 1195105 * 4/2002
EP 1563759 * 8/2005
FR 2866215 * 8/2005

(Continued)

OTHER PUBLICATIONS

Machine Translation of FR 2866215; Available Aug. 2005.*

(Continued)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 872 days.

(21) Appl. No.: **17/272,183**

(22) PCT Filed: **Aug. 29, 2019**

(86) PCT No.: **PCT/EP2019/073071**

§ 371 (c)(1),

(2) Date: **Feb. 26, 2021**

(87) PCT Pub. No.: **WO2020/043816**

PCT Pub. Date: **Mar. 5, 2020**

(65) **Prior Publication Data**

US 2021/0307487 A1 Oct. 7, 2021

(30) **Foreign Application Priority Data**

Aug. 29, 2018 (FR) 1857744

(51) **Int. Cl.**

A45D 34/04 (2006.01)

A45D 40/20 (2006.01)

(52) **U.S. Cl.**

CPC **A45D 34/045** (2013.01); **A45D 40/20** (2013.01); **A45D 2200/1072** (2013.01)

(58) **Field of Classification Search**

CPC **A45D 34/045**; **A45D 40/265**

See application file for complete search history.

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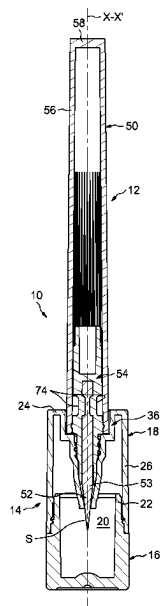
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ABSTRACT

The applicator for applying a product, in particular a cosmetic product, comprises a gripping cap (50) and an applicator member (52) that is provided with at least one application surface for said product and comprises a flexible portion (53). The applicator also comprises a rigid assembly insert (54) that is mounted at least partially inside the gripping cap (50) and is secured to said cap. The flexible portion (53) of the applicator member is fastened in the assembly insert (54) in a region of said insert which is mounted inside the gripping cap (50).

20 Claims, 3 Drawing Sheets



(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	2008-006267	A		1/2008
JP	20086267		*	1/2008
JP	2009-034145	A		2/2009

OTHER PUBLICATIONS

Espacenet Machine Translation of EP 1563759; Available Aug. 2005.*

Espacenet Machine Translation of JP 20086267; Available Jan. 2008.*

Machine Translation of EP 1195105; Available Apr. 2002.*

* cited by examiner

FIG.1

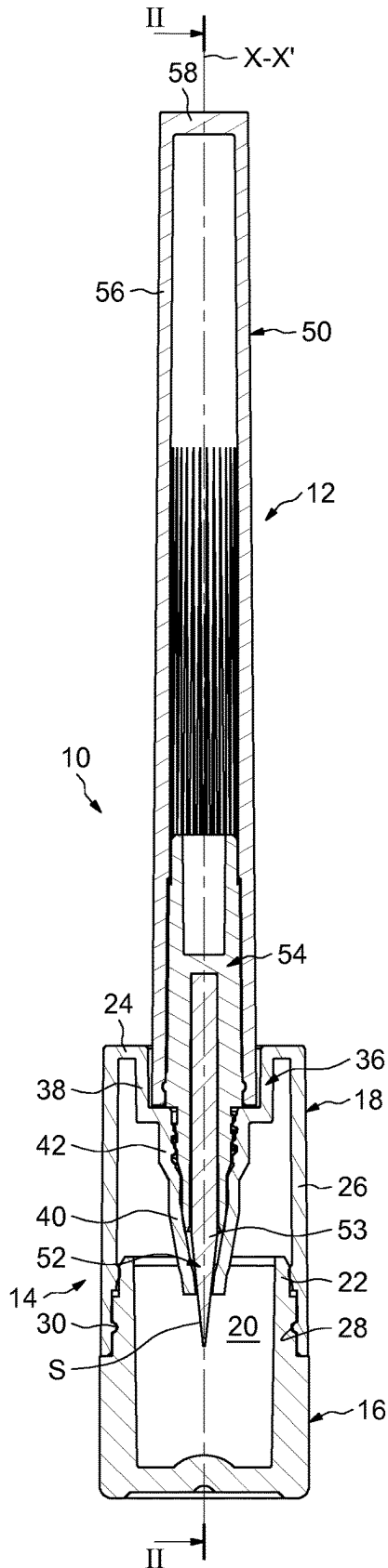


FIG.2

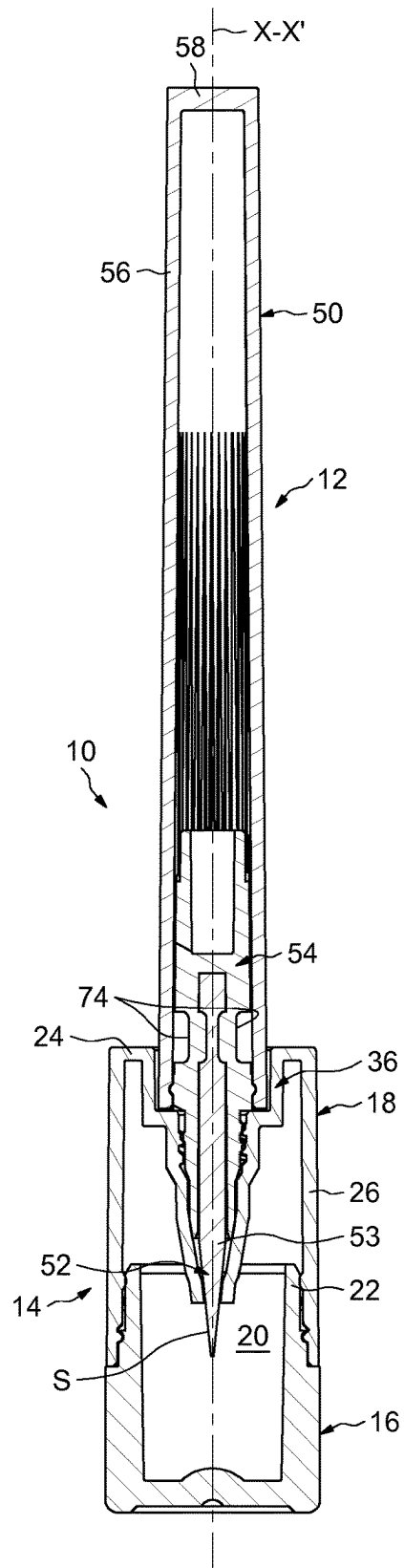


FIG.3

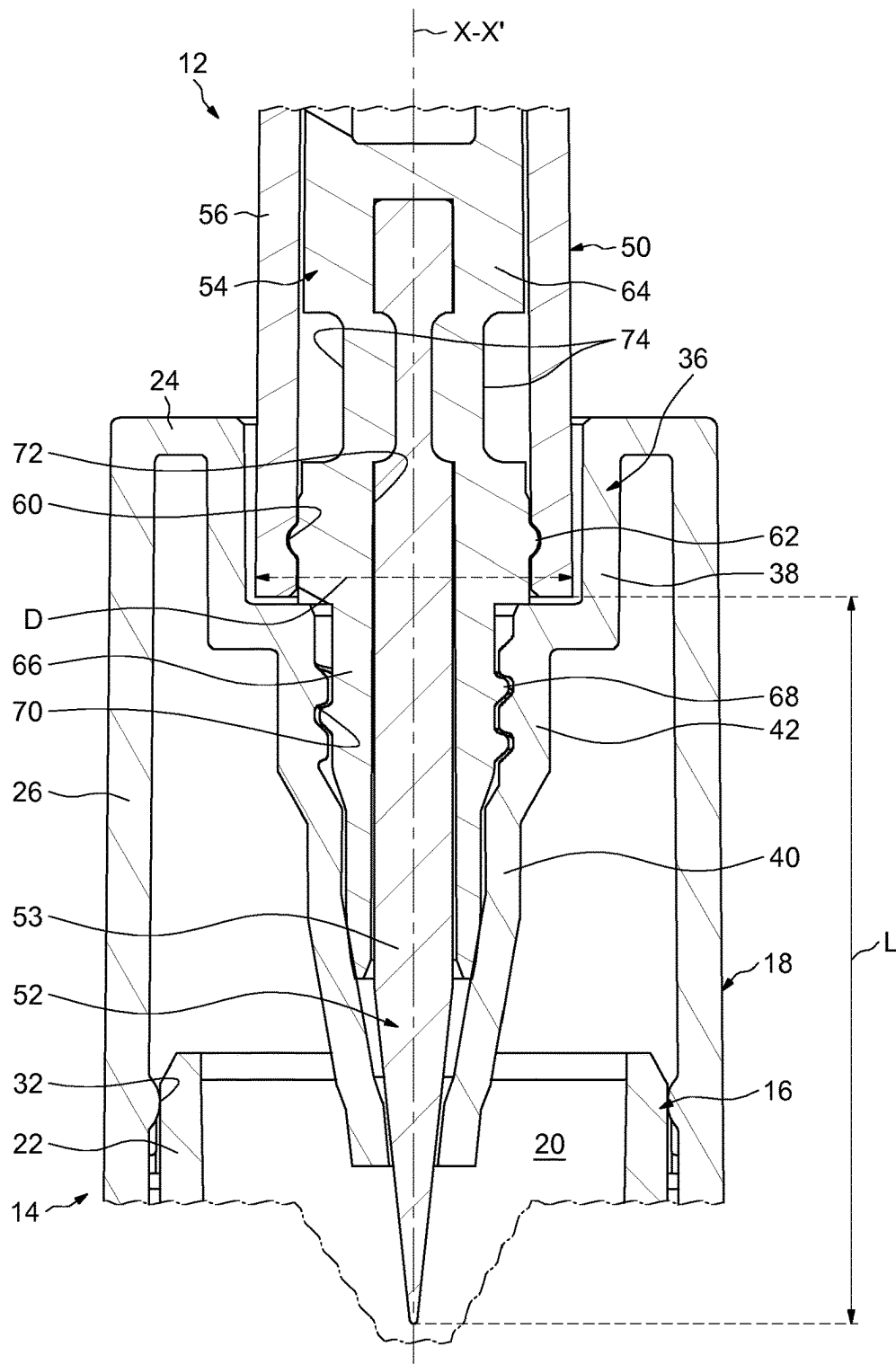


FIG.4

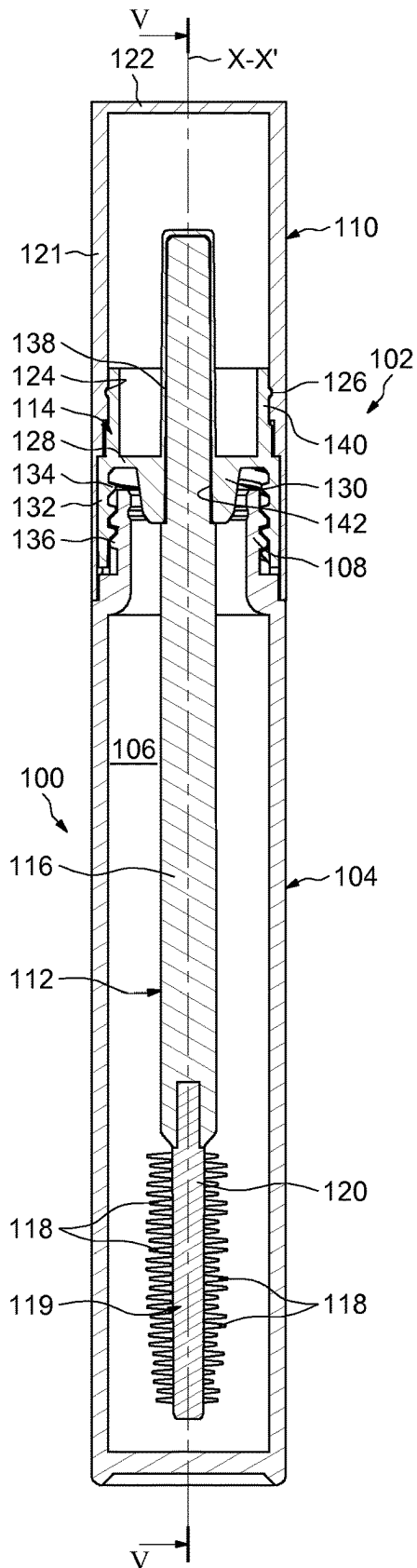
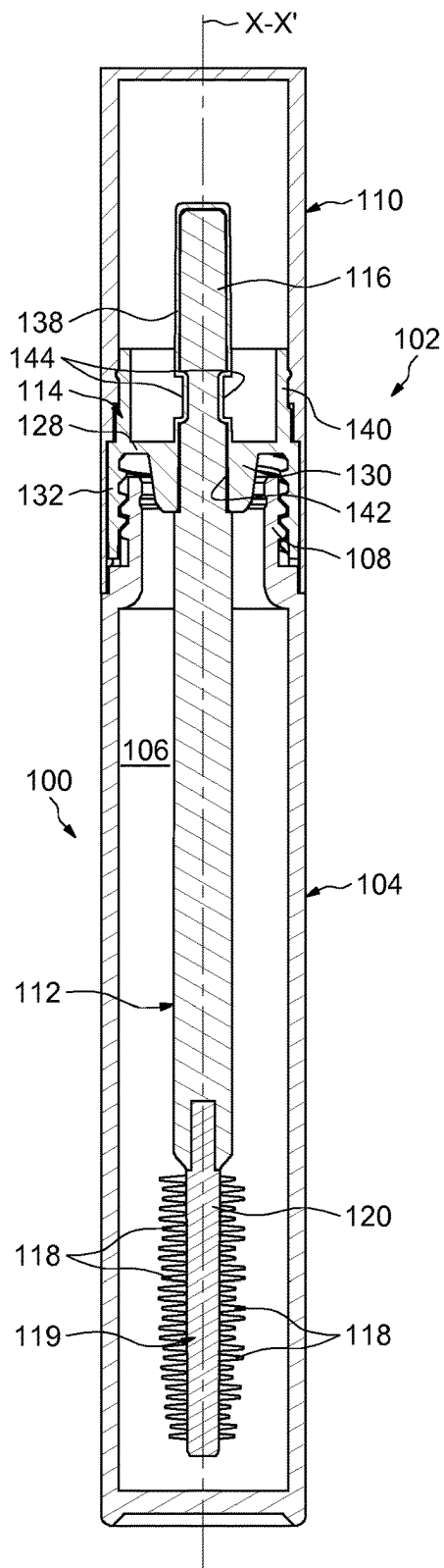


FIG.5



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APPLICATOR FOR APPLYING A PRODUCT, IN PARTICULAR A COSMETIC PRODUCT

The present invention relates to an applicator for applying a product, in particular a cosmetic product.

The expression “cosmetic product” is understood to mean a product as defined in Regulation (EC) No 1223/2009 of the European Parliament and of the Council of 30 Nov. 2009 relating to cosmetic products.

The invention is particularly designed for the application of a cosmetic product, such as a makeup product, for example an eyeliner in particular for drawing a line on the eyelids, or mascara to be applied to the eyelashes or eyebrows.

For the application of these types of product, use is conventionally made of an applicator of the type comprising a gripping cap, a protruding stem secured to the cap, and an applicator member fastened to the free end of the stem. The applicator member may for example be in the form of an applicator tip for applying an eyeliner, or in the form of a brush or comb attached to the stem for applying mascara. For more details on the design of such an applicator, reference may be made for example to the Patent Application WO-A2-2011/004341.

With this type of applicator, it may be difficult to obtain precise application of the product, in particular when it is an eyeliner to be deposited in the form of a line, given the significant axial distance between the free end of the applicator member and the gripping cap. Moreover, this does not make it possible to obtain entirely satisfactory use ergonomics.

The present invention aims to overcome these drawbacks.

An object of the invention is an applicator for applying a product, in particular a cosmetic product, comprising a gripping cap and an applicator member that is provided with at least one application surface for said product and comprises a flexible portion.

According to a first general feature, the applicator also comprises a rigid assembly insert that is mounted at least partially inside the gripping cap and is secured to said cap.

According to a second general feature, the flexible portion of the applicator member is fastened in the assembly insert in a region of said insert which is mounted inside the gripping cap.

In contrast to “rigid assembly insert”, a “flexible portion of the applicator member” is understood to be a portion of the applicator member that is capable, on account of the material used and/or the dimensioning thereof, of deforming under the action of a load exerted by the user following contact with the surface to be treated, and of returning by elasticity to its initial position once the load ceases.

Fastening the flexible portion of the applicator member in a region of the assembly insert that is disposed axially inside the gripping cap makes it possible to reduce the axial distance between this cap and the free end of the applicator member.

Thus, the gripping region of the cap is closer to the applicator member compared with prior art applicators. This encourages good use ergonomics and improved precision of application. The overall axial size of the applicator is also reduced.

Moreover, the fastening region(s) between the assembly insert and the flexible portion of the applicator member that may be visible at the outer surface of said insert are concealed by the presence of the gripping cap around this or

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these fastening region(s). This contributes towards the appearance and the perceived quality of the applicator to the user.

Furthermore, the flexibility of the flexible portion of the applicator member allows it to deform, for example by bending, reversibly and thereby to provide a degree of comfort during application.

According to one advantageous provision, the flexible portion of the applicator member is fastened in the assembly insert by local plastic deformation of said insert.

The rigid nature of the assembly insert makes it possible to secure it to the flexible portion of the applicator member in a simple, safe and economic manner. The local plastic deformation(s) can be obtained for example by material creep, for example by crimping or by heading. Alternatively, it is possible to join the flexible portion of the applicator member to the assembly insert by other methods, for example by ultrasonic welding.

Preferably, the flexible portion of the applicator member is made of flexible material and the assembly insert is made of rigid material.

According to one particular provision, the Shore D hardness of the assembly insert may be greater than the Shore D hardness of the flexible portion of the applicator member by at least 20 units of hardness. The Shore D hardness of the flexible portion of the applicator member may for example be less than 60 and greater than 20.

According to one particular design, the gripping cap, the assembly insert and the applicator member may be designed such that the diameter D of the free end of said cap is between 8 mm and 12 mm and the axial length L between the free end of said cap and the free end of the applicator member is between 10 mm and 25 mm, with in particular an L/D ratio of between 0.83 and 3.13.

This particular design aims to make it easier to hold the gripping cap in the hand and to optimize the axial distance between the cap and the free end of the applicator member.

The assembly insert is preferably provided with a housing, inside which the flexible portion of the applicator member is mounted and fastened.

Advantageously, the assembly insert comprises at least one thread for making it possible to fasten said applicator to a container. The thread may be helical in order to allow fastening by screwing. Alternatively, it is possible to provide on the assembly insert one or more reliefs that make it possible to fasten it to the container by snap-fitting or snap-fastening. In this case, each relief forms coupling or clip-fastening teeth.

Thus, the applicator is fastened to the container by the assembly insert and not by the gripping cap as is the case in the prior art applicators.

In order to axially secure the gripping cap relative to the assembly insert, the latter may comprise at least one retaining member that engages with at least one complementary retaining member on the gripping cap.

In one embodiment, the applicator member comprises an applicator tip that forms said flexible portion and defines said product application surface. In an alternative embodiment, the applicator member may comprise a stem that forms said flexible portion and a plurality of application elements that are carried by the stem and each define a product application surface.

The invention also relates to a device for packaging and applying a product, in particular a cosmetic product, comprising a container for storing said product, and an applicator as defined above that is mounted in a removable manner between a closing position in which the assembly insert

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seals off the storage container and the applicator member is situated inside said container, being at least partially in contact with the product contained therein, and an application position in which the applicator is separated from said container.

In one embodiment, the storage container comprises a body and a collar which is fastened to the neck of the body and on which the assembly insert of the applicator is mounted in a removable manner. The collar may comprise an internal skirt, inside which the assembly insert and an end portion of the gripping cap of the applicator are fitted when the latter is in the position closing the storage container. This internal skirt may for example be provided with a fastening portion comprising at least one thread designed to engage with a thread of the assembly insert, and with a sealing portion in sealing contact with said insert, when the applicator is in the position closing the storage container.

In another embodiment, the assembly insert of the applicator is fastened to the neck of the storage container.

The present invention will be understood better from studying the detailed description of embodiments that are given by way of entirely non-limiting example and are illustrated by the appended drawings, in which:

FIG. 1 is a view in longitudinal section of a packaging and application device comprising an applicator according to a first exemplary embodiment of the invention,

FIG. 2 is a view in cross section on the axis II-II in FIG. 1,

FIG. 3 is a detail view of FIG. 2,

FIG. 4 is a view in longitudinal section of a packaging and application device comprising an applicator according to a second exemplary embodiment of the invention, and

FIG. 5 is a view in cross section on the axis V-V in FIG. 4.

FIGS. 1 and 2 show a device, denoted by the overall reference numeral 10, which is intended to allow the packaging and application of a cosmetic product. The cosmetic product to be applied may be for example an eyeliner.

The device 10, of longitudinal axis X-X', is shown in a position that is presumed to be vertical. The device 10 comprises a product applicator 12 and a container 14 containing the cosmetic product to be applied (not shown).

As will be described in more detail below, the applicator 12 is designed to achieve in particular a good degree of comfort during application of the product and improved precision of application.

In the exemplary embodiment illustrated, the container 14 comprises a body 16, of axis X-X', and a collar 18 fastened to the body. The body 16 internally delimits an internal volume or reservoir 20 for storing the cosmetic product. The body 16 comprises an end forming a bottom and an open opposite end forming a neck 22 that delimits an opening for accessing the internal reservoir 20. In the exemplary embodiment illustrated, the body 16 has a circular cross section. In a variant, the body 16 may have a cross section that is polygonal, in particular square, or an oval cross section. The body 16 is made in one piece by moulding a synthetic material. Alternatively, the body may be made of other materials, for example of glass. With such a material, a seal may be provided between the body 16 and the collar 18.

The collar 18 makes it possible in particular to removably mount the applicator 12 on the container 14. The collar 18 is likewise made in one piece by moulding a synthetic material. The collar 18 comprises an upper frontal end wall 24 and a peripheral external fastening skirt 26 extending said wall so as to allow the collar to be fastened to the body 16.

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The fastening skirt 26 is centred on the axis X-X'. The fastening skirt 26 extends axially from the end wall 24. The end wall 24 extends radially. The fastening skirt 26 axially continues a large-diameter edge of the end wall 24. The skirt 26 has a circular cross section.

The fastening skirt 26 of the collar comprises, on its inner surface, a bulge 28 for retaining the body 16 that extends inwards and engages by snap-fastening in a corresponding groove 30 formed in the outer surface of the neck 22 of said body. The bulge 28 may be continuous or discontinuous in the circumferential direction. As illustrated more clearly in FIG. 3, the fastening skirt 26 also comprises, on its inner surface, an annular sealing rib 32 that extends inwards. The rib 32 bears in a sealing manner against the neck 22 of the body. The rib 32 is situated axially above the snap-25 fastening bulge 28.

The fastening skirt 26 may also comprise, on its inner surface, a plurality of axial fins that are spaced apart from one another in the circumferential direction and engage with fins provided on the body 16 in order to provide an anti-rotation function preventing the collar 18 from rotating relative to said body.

The collar 18 comprises an internal annular mounting skirt 36 for accommodating the applicator 12 on and fastening it to the container 14, as will be described in more detail below. The mounting skirt 36 extends axially from the frontal end wall 24. The skirt 36 is centred on the axis X-X'. The external fastening skirt 26 radially surrounds the internal mounting skirt 36, remaining radially at a distance therefrom.

The internal mounting skirt 36 has a stepped shape. The internal skirt 36 is provided with a proximal mounting portion 38 extending from the frontal end wall 24, with a distal sealing portion 40 and with a fastening portion 42 extending between said mounting portion 38 and sealing portion 40. The fastening portion 42 has a smaller outside diameter than the outside diameter of the mounting portion 38, thereby leaving a radial shoulder (not referenced) therebetween. The sealing portion 40 axially continues the fastening portion 42 on the opposite side from the mounting portion 38. In the exemplary embodiment illustrated, the sealing portion 40 has an upwardly diverging conical part.

Referring again to FIGS. 1 and 2, the applicator 12 comprises a gripping cap 50, an applicator member 52 made in this case in the form of an applicator tip 53, and an assembly insert 54 that is fastened to the cap 50 and bears the applicator tip 53. The insert 54 is separate from the cap 50 and connected thereto. The insert 54 is secured to the cap 50. The insert 54 is also separate from the applicator member 52.

The applicator tip 53 is made of flexible material, in particular of flexible synthetic material. The applicator tip 53 may be for example a foam, a felt, a rubber, or a flocked body. The free end part of the applicator tip 53 defines a product application surface S that is intended to come into contact with the area of the user's body to be treated. The application surface S has in this case a frustoconical shape so as to promote the precision of application of the product. Alternatively, however, it remains possible to provide other shapes.

The cap 50, of axis X-X', comprises a tubular peripheral wall 56 and an upper frontal end wall 58 from which the wall 56 extends. In the exemplary embodiment illustrated, the peripheral wall 56 has a circular cross section. In a variant, the peripheral wall 56 may have a cross section that is polygonal, in particular square, or an oval cross section.

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In order to fasten the cap **50** to the insert **54**, the peripheral wall **56** of said cap comprises, on its inner surface, a groove **60** in which a retaining bulge **62** formed on the outer surface of the insert **54** engages by snap-fastening, as is illustrated more clearly in FIG. 3. The bulge **62** extends radially outwards. The bulge **62** may be continuous or discontinuous in the circumferential direction.

The insert **54** is made of rigid material. The insert may be made for example of rigid synthetic material, for example of PP, PE, etc., or of steel or aluminium.

In the exemplary embodiment illustrated, the insert **54** is mounted partially inside the cap **50** and protrudes axially out of said cap. The insert **54** comprises a proximal fastening portion **64** and a distal mounting portion **66**. The fastening portion **64** makes it possible to fasten the cap **50** to the insert **54** and to fasten the applicator tip **53** inside said insert. The mounting portion **66** makes it possible to removably mount the applicator **12** on the container **14**.

The fastening portion **64** of the insert is equipped, on its outer surface, with the bulge **62** for fastening the cap **50** to the insert. The fastening portion **64** is mounted inside the cap **50**, and more specifically inside the peripheral wall **56** of said cap. The cap **50** covers the fastening portion **64** of the insert. The cap **50** is mounted on the insert **54** simply by being pushed axially. To make this mounting easier, a slight radial clearance may be provided between the outer surface of the fastening portion **64** of the insert and the inner surface of the peripheral wall **56** of the cap.

The mounting portion **66** of the insert axially continues the fastening portion **64**. The mounting portion **66** protrudes axially out of the cap **50**. The mounting portion **66** has a smaller outside diameter than the outside diameter of the fastening portion **64**. A radial shoulder (not referenced) is formed between these portions and in this case comes to bear axially against the radial shoulder formed on the skirt **36** of the collar **18** of the container. The end part of the cap **50** that covers the insert **54** extends partially inside the mounting portion **38** of the skirt.

The mounting portion **66** of the insert comprises, on its outer surface, a helical thread **68** engaged with a corresponding thread **70** formed on the skirt **36** of the collar **18** of the container for screwing the applicator **12** to the container **14**. The thread **70** is provided on the inner surface of the fastening portion **42** of the skirt. The threads **68**, **70** may be continuous or discontinuous in the circumferential direction. The mounting portion **66** comes into sealing contact in the radial direction with the sealing portion **40** of the skirt **36** when the applicator **12** is in the position closing the container **14**.

The insert **54** comprises a housing **72** inside which the applicator tip **53** is mounted and fastened. The housing **72** extends axially from the free end of the mounting portion **66** to the fastening portion **64**. The housing **72** is centred on the axis X-X'. The housing **72** is made in this case in the form of a blind hole. The applicator tip **53** protrudes axially out of the insert **54**. The applicator tip **53** thus protrudes out of the housing **72**. The applicator tip **53** extends without contact inside the sealing portion **40** of the skirt **36** of the collar **18**. The applicator tip **53** protrudes axially out of said sealing portion **40**.

The applicator tip **53** is fastened inside the insert by local plastic deformations of the insert. These deformations, referenced **74**, of the insert are visible in FIGS. 2 and 3. The local plastic deformations **74** are realized on the fastening portion **64** of the insert in the region of the housing **72**. The deformations **74** are formed by material being embossed radially without chips being removed so as to locally bring

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about material creep radially towards the inside against the applicator tip **53**. The flexible applicator tip **53** is also locally deformed. The applicator tip **53** is thus fastened inside the housing **72**. This operation may be realized for example by crimping or by heading, and is carried out before the cap **50** is mounted on the insert **54**.

In the closing position illustrated in the figures, in which the applicator **12** is mounted on the container **14**, the insert **54** closes off the collar **18** of the container and the applicator tip **53** is situated inside the internal reservoir **20** of said container such that it can be filled, for example by contact or by capillarity, with product to be applied. When the user wishes to apply the product, they take hold of the cap **50** of the applicator **12**, unscrew the insert **54** by rotating the cap, then handle said cap to obtain the desired application.

Given that the applicator tip **53** is fastened to a part of the insert **54** that is covered by the cap **50**, this cap can be gripped by the user at a small axial distance from the free end of the applicator tip **53**. This favours the ergonomics of use and also allows good precision of application. The flexibility of the applicator tip **53** also provides a degree of comfort during application and makes this application easier.

Advantageously, the diameter D of the free end of the cap **50** is between 8 mm and 12 mm and the axial length L between the free end of the cap **50** and the free end of the applicator tip **53** is between 10 mm and 25 mm, with in particular an L/D ratio of between 0.83 and 3.13.

With such an L/D ratio, the user has an applicator **10** of which the cap **50** has a diameter that makes it easier to grip, and having an applicator tip **53** end situated as close as possible to the gripping part of the cap in order to further improve the ergonomics and the precision of application.

In this first exemplary embodiment, the applicator of the device **10** is provided to apply a cosmetic product of the eyeliner type. It is possible to provide for the applicator that is the subject of the invention to be used to apply other types of cosmetic product.

For example, in the exemplary embodiment illustrated in FIGS. 4 and 5, the device **100** is intended to allow the packaging and application of a cosmetic product of the mascara type.

The device **100**, of longitudinal axis X-X', comprises a product applicator **102** and a container **104** containing the cosmetic product to be applied (not shown).

In this exemplary embodiment, the container **104** is made in one piece. The container **104** internally delimits an internal volume or reservoir **106** for storing the cosmetic product. The container **104** comprises an end forming a bottom and an open opposite end forming a neck **108** that delimits an opening for accessing the internal reservoir **106**.

The applicator **102** comprises a gripping cap **110**, an applicator member **112**, and an assembly insert **114** that is fastened to the cap **110** and bears the applicator member **112**. The insert **114** is separate from the cap **110** and connected thereto. The insert **114** is secured to the cap **110**. The insert **114** is also separate from the applicator member **112**.

The applicator member **112** comprises a stem **116** extending along the axis X-X' and a plurality of application elements **118** borne by the stem **116**. The applicator member **112** comprises an applicator end piece **119** that is fastened to the stem **116** and comprises the application elements **118**. The application elements **118** are borne by the stem **116** via the applicator end piece **119**. The application elements **118** protrude from the applicator end piece **119**.

Each application element **118** defines a product application surface intended to come into contact with the user's keratin fibres, in this case the eyelashes and/or eyebrows.

The stem **116** is made of flexible material, in particular of flexible synthetic material. By way of indication, the stem **116** may for example be made of a flexible synthetic material such as SEBS, a silicone, latex, butyl, EPDM, a nitrile, a thermoplastic elastomer, etc. The stem **116** in this case has a circular cross section. In a variant, the stem **116** may have a cross section that is polygonal, in particular square, or an oval cross section, etc.

The application elements **118** are mounted on the applicator end piece **119**. The application elements **118** have a function of separating and combing the user's keratin fibres, and delimit spaces between one another for these fibres to pass through. Each application element **118** extends radially outwards from the applicator end piece **119**.

The application elements **118** are produced in this case in the form of spikes. The application elements **118** may have other forms, for example bristles, and/or teeth, and/or points, and/or striations, and/or bosses, and/or fins, etc. By way of indication, the height of the application elements **118** may be between 1 and 6 mm. The application elements **118** may have an equal height or have a height that varies along the axis X-X'. It is also possible to provide application elements that differ from one another by way of at least one of their shape, thickness, height, orientation, colour and/or material.

The applicator end piece **119** and the application elements **118** may for example be moulded from one and the same synthetic material, or in a variant they can be made from at least two different synthetic materials. The application elements **118** may for example be produced in one piece by moulding together with the applicator end piece **119** or by being overmoulded on said stem, or be fastened to the stem by any appropriate means, for example by adhesive bonding, flocking, crimping, etc.

The applicator end piece **119**, of axis X-X', is mounted at the free end of the stem **116**. To this end, a blind cutout (not referenced) is made in the lower end of the stem **116**, inside which the applicator end piece **119** is partially accommodated. The applicator end piece **119** may be fastened to the stem **116** by any appropriate means, for example by wedging, screw fastening, welding, adhesive bonding, crimping, etc. The applicator end piece **119** comprises a core **120** of elongate shape, of axis X-X', on which the application elements **118** are disposed.

The cap **110**, of axis X-X', comprises a tubular peripheral wall **121** and an upper frontal end wall **122** from which the wall **121** extends. In order to fasten the cap **110** to the insert **114**, the peripheral wall **112** of said cap comprises, on its inner surface, a groove **124** in which a retaining bulge **126** formed on the outer surface of the insert **114** engages by snap-fastening. The bulge **126** extends radially outwards. The bulge **126** may be continuous or discontinuous in the circumferential direction.

The insert **114** is made of rigid material. The insert may be made for example of rigid synthetic material, for example of PP, PE, etc., or of steel or aluminium. In this illustrated exemplary embodiment, the insert **114** is mounted entirely inside the cap **110**.

The insert **114** comprises an upper frontal end wall **128**, a first internal mounting skirt **130** and a first peripheral external fastening skirt **132** surrounding the internal skirt. The skirts **130**, **132** are centred on the axis X-X'. The skirts **130**, **132** extend axially from the frontal wall **128**. The frontal wall **128** extends radially.

The external skirt **132** axially continues a large-diameter edge of the frontal wall **128**. The external skirt **132** radially surrounds the internal skirt **130**, remaining radially at a distance from the latter. The external skirt **132** in this case extends axially beyond the internal skirt **130**. The skirt **132** has a circular cross section.

The external skirt **132** internally comprises a helical thread **134** for screwing the insert **114** onto a corresponding thread **136** formed on the neck **108** of the container. The threads **134**, **136** may be continuous or discontinuous in the circumferential direction.

The insert **114** also comprises a second internal mounting skirt **138** and a second peripheral external fastening skirt **140** surrounding the internal skirt, remaining radially at a distance therefrom. The skirts **138**, **140** extend axially from the frontal wall **128** on the opposite side from the skirts **130**, **132**. The skirts **138**, **140** are centred on the axis X-X'.

The external fastening skirt **140** is equipped, on its outer surface, with the bulge **126** for fastening the cap **110** to the insert **114**. The mounting skirt **138** is situated in the axial continuation of the mounting skirt **130**.

The insert **114** comprises a housing **142** which is delimited by the mounting skirts **130**, **138** and inside which the stem **116** of the applicator member **112** is mounted and fastened. The housing **142** extends axially from the free end of the skirt **130** to a bottom wall of the skirt **138**.

In a similar manner to the first example, the stem **116** is fastened inside the insert **114** by local plastic deformations of the insert. These deformations, referenced **144**, of the insert are visible in FIG. 5. The local plastic deformations **144** are realized on the mounting skirt **138**. The flexible stem **116** of the applicator member **112** is also deformed locally. The stem **116** is fastened inside the housing **142**. The fastening may be realized for example by crimping or by heading, and is carried out before the cap **110** is mounted on the insert **114**.

By virtue of the invention, an applicator is provided that has a limited axial size and good use ergonomics while making it possible to achieve improved precision of application.

The invention claimed is:

1. An applicator for applying a product, comprising a gripping cap, an applicator member that is provided with at least one application surface for said product and comprises a flexible portion, and a rigid assembly insert that is mounted at least partially inside the gripping cap and is secured to said cap,

wherein the assembly insert comprises at least one fastening element that is able to allow said applicator to be fastened to a container,

wherein the gripping cap is mounted on an outer surface of the assembly insert,

wherein the flexible portion of the applicator member is fastened in the assembly insert in a region of said insert that is mounted inside the gripping cap,

wherein the flexible portion of the applicator member is fastened in the assembly insert by local plastic deformations of the outer surface of said insert which extend radially against the flexible portion of the applicator member,

wherein said local plastic deformations are visible at the outer surface of the assembly insert and concealed by the gripping cap,

wherein the assembly insert comprises a first portion, a second portion comprising the local plastic deformations distal the first portion, and a third portion abutting the gripping cap distal the second portion,

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wherein the first portion, second portion, and third portion are each mounted inside the gripping cap, wherein the second portion defines at least one groove around an outer circumference of the outer surface of the assembly insert, said groove being formed by said local plastic deformations.

2. The applicator for applying the product according to claim 1, wherein the flexible portion of the applicator member is fastened in the assembly insert by local plastic deformation of said insert by crimping.

3. The applicator for applying the product according to claim 1, wherein the assembly insert is provided with a housing, inside which the flexible portion of the applicator member is mounted and fastened.

4. The applicator for applying the product according to claim 1, wherein the flexible portion of the applicator member is made of flexible material and the assembly insert is made of rigid material.

5. The applicator for applying the product according to claim 1, wherein the Shore D hardness of the assembly insert is greater than the Shore D hardness of the flexible portion of the applicator member by at least 20 units of hardness.

6. The applicator for applying the product according to claim 1, wherein the gripping cap, the assembly insert and the applicator member are designed such that a diameter D of said cap is between 8 mm and 12 mm and an axial length L between said cap and the applicator member is between 10 mm and 25 mm, with in particular an L/D ratio of between 0.83 and 3.13.

7. The applicator for applying the product according to claim 1, wherein the assembly insert comprises at least one thread that is able to allow said applicator to be fastened to the container.

8. The applicator for applying the product according to claim 1, wherein the assembly insert comprises at least one retaining member that engages with at least one complementary retaining member of the gripping cap in order to axially secure said cap relative to the assembly insert.

9. The applicator for applying the product according to claim 1, wherein the applicator member comprises a stem that forms said flexible portion and a plurality of application elements that are carried by the stem and each define a product application surface.

10. The applicator for applying the product of claim 1, wherein a cross section of the at least one groove is a rounded rectangle.

11. The applicator for applying the product of claim 1, wherein the at least one groove extends circumferentially around the outer surface of the assembly insert.

12. The applicator for applying the product of claim 1, wherein the first portion comprises a proximal fastening portion.

13. The applicator for applying the product of claim 1, wherein the third portion comprises a bulge.

14. A device for packaging and applying a product, in particular a cosmetic product, comprising a storage container, for storing said product, a neck of the storage container, and an applicator according to claim 1 that is mounted in a removable manner between a closing position in which the assembly insert seals off the storage container and the applicator member is situated inside said storage container, being at least partially in contact with the product contained therein, and an application position in which the applicator is separated from said storage container.

15. The device for packaging and applying the product according to claim 14, wherein the storage container comprises a body and a collar which is fastened to the neck of

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the storage container and on which the assembly insert of the applicator is mounted in a removable manner.

16. The device for packaging and applying the product according to claim 15, wherein the collar comprises an internal skirt, inside which the assembly insert and an end portion of the gripping cap of the applicator are fitted.

17. The device for packaging and applying the product according to claim 16, wherein the internal skirt of the collar is provided with a fastening portion comprising at least one thread engaged with a thread of the assembly insert, and with a sealing portion in sealing contact with said insert.

18. The device for packaging and applying the product according to claim 14, wherein the assembly insert of the applicator is fastened to the neck of the storage container.

19. An applicator for applying a product, comprising a gripping cap, an applicator member that is provided with at least one application surface for said product and comprises a flexible portion, and a rigid assembly insert that is mounted at least partially inside the gripping cap and is secured to said cap,

wherein the assembly insert comprises at least one fastening element that is able to allow said applicator to be fastened to a container,

wherein the gripping cap comprises an upper end wall and a peripheral wall extending from the upper end wall, wherein the flexible portion of the applicator member protrudes axially out of the assembly insert and is fastened in the assembly insert in a region of said insert that is mounted inside the gripping cap,

wherein the flexible portion of the applicator member is fastened in the assembly insert by local plastic deformations of said insert which extend radially against the flexible portion of the applicator member,

wherein said local plastic deformations are visible at the assembly insert and concealed by the gripping cap, wherein fastening regions between the assembly insert and the flexible portion of the applicator member are axially offset toward the upper end wall of the gripping cap with respect to the fastening element of the assembly insert,

wherein the assembly insert comprises a first portion, a second portion comprising the local plastic deformations distal the first portion, and a third portion abutting the gripping cap distal the second portion,

wherein the first portion, second portion, and third portion are each mounted inside the gripping cap,

wherein the second portion defines at least one groove around an outer circumference of an outer surface of the assembly insert, said groove being formed by said local plastic deformations.

20. An applicator for applying a product, comprising a gripping cap, an applicator member that is provided with at least one application surface for said product and comprises a flexible portion, and a rigid assembly insert that is mounted at least partially inside the gripping cap and is secured to said cap,

wherein the assembly insert comprises at least one fastening element that is able to allow said applicator to be fastened to a container,

wherein the gripping cap is mounted on an outer surface of the assembly insert,

wherein the flexible portion of the applicator member is fastened in the assembly insert in a region of said insert that is mounted inside the gripping cap,

wherein the flexible portion of the applicator member is fastened in the assembly insert by local plastic deformations of the outer surface of said insert,

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wherein a radial space is formed between the gripping cap
and each local plastic deformation of the outer surface
of the assembly insert,
wherein said local plastic deformations are visible at the
outer surface of the assembly insert and concealed by 5
the gripping cap,
wherein the assembly insert comprises a first portion, a
second portion comprising the local plastic deforma-
tions distal the first portion, and a third portion abutting
the gripping cap distal the second portion, 10
wherein the first portion, second portion, and third portion
are each mounted inside the gripping cap,
wherein the second portion defines at least one groove
around an outer circumference of the outer surface of
the assembly insert, said groove being formed by said 15
local plastic deformations.

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